Thomas Algorithm

Thomas Algorithm

Tridiagonal System:

$$\begin{bmatrix} b_1 & c_1 & & & \\ a_2 & b_2 & c_2 & & \\ & a_3 & b_3 & c_3 & \\ & & \ddots & \ddots & \ddots \\ & & & a_n & b_n \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{Bmatrix} = \begin{Bmatrix} d_1 \\ d_2 \\ d_3 \\ \vdots \\ d_n \end{Bmatrix}$$

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Tridiagonal System:

$$\begin{bmatrix} b_1 & c_1 & & & \\ a_2 & b_2 & c_2 & & \\ & a_3 & b_3 & c_3 & \\ & & \ddots & \ddots & \ddots \\ & & & a_n & b_n \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{Bmatrix} = \begin{Bmatrix} d_1 \\ d_2 \\ d_3 \\ \vdots \\ d_n \end{Bmatrix}$$

Forward Elimination:

$$\begin{bmatrix} b_1 & c_1 & & & \\ & b_2' & c_2 & & \\ & a_3 & b_3 & c_3 & \\ & & \ddots & \ddots & \ddots \\ & & & a_n & b_n \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{Bmatrix} = \begin{Bmatrix} d_1 \\ d_2' \\ d_3 \\ \vdots \\ d_n \end{Bmatrix}$$

Thomas Algorithm

Tridiagonal System:

$$\begin{bmatrix} b_1 & c_1 & & & \\ a_2 & b_2 & c_2 & & \\ & a_3 & b_3 & c_3 & \\ & & \ddots & \ddots & \ddots \\ & & & a_n & b_n \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{Bmatrix} = \begin{Bmatrix} d_1 \\ d_2 \\ d_3 \\ \vdots \\ d_n \end{Bmatrix}$$

Forward Elimination:

$$\begin{bmatrix} b_1 & c_1 & & & \\ & b_2' & c_2 & & \\ & a_3 & b_3 & c_3 & \\ & & \ddots & \ddots & \ddots \\ & & & a_n & b_n \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{Bmatrix} = \begin{Bmatrix} d_1 \\ d_2' \\ d_3 \\ \vdots \\ d_n \end{Bmatrix}$$

$$b_2' = b_2 - \frac{a_2}{b_1} \cdot c_1$$

$$d_2' = d_2 - \frac{a_2}{b_1} \cdot d_1$$

Thomas Algorithm

Tridiagonal System:

$$\begin{bmatrix} b_1 & c_1 & & & \\ a_2 & b_2 & c_2 & & \\ & a_3 & b_3 & c_3 & \\ & & \ddots & \ddots & \ddots \\ & & & a_n & b_n \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{Bmatrix} = \begin{Bmatrix} d_1 \\ d_2 \\ d_3 \\ \vdots \\ d_n \end{Bmatrix}$$

Forward Elimination:

$$\begin{bmatrix} b_1 & c_1 & & & \\ & b_2' & c_2 & & \\ & a_3 & b_3 & c_3 & \\ & & \ddots & \ddots & \ddots \\ & & & a_n & b_n \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{Bmatrix} = \begin{Bmatrix} d_1 \\ d_2' \\ d_3 \\ \vdots \\ d_n \end{Bmatrix}$$

$$b_2' = b_2 - \frac{a_2}{b_1} \cdot c_1$$

$$d_2' = d_2 - \frac{a_2}{b_1} \cdot d_1$$

$$\begin{bmatrix} b_1 & c_1 & & & \\ & b_2' & c_2 & & \\ & & b_3' & c_3 & \\ & & & \ddots & \ddots \\ & & & & b_n' \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{Bmatrix} = \begin{Bmatrix} d_1 \\ d_2' \\ d_3'' \\ \vdots \\ d_n^{(n-1)} \end{Bmatrix}$$

Thomas Algorithm

Tridiagonal System:

$$\begin{bmatrix} b_1 & c_1 & & & \\ a_2 & b_2 & c_2 & & \\ & a_3 & b_3 & c_3 & \\ & & \ddots & \ddots & \ddots \\ & & & a_n & b_n \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{Bmatrix} = \begin{Bmatrix} d_1 \\ d_2 \\ d_3 \\ \vdots \\ d_n \end{Bmatrix}$$

Forward Elimination:

$$\begin{bmatrix} b_1 & c_1 & & & \\ & b_2' & c_2 & & \\ & a_3 & b_3 & c_3 & \\ & & \ddots & \ddots & \ddots \\ & & & a_n & b_n \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{Bmatrix} = \begin{Bmatrix} d_1 \\ d_2' \\ d_3 \\ \vdots \\ d_n \end{Bmatrix}$$

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$$d_2' = d_2 - \frac{a_2}{b_1} \cdot d_1$$

$$\begin{bmatrix} b_1 & c_1 & & & \\ & b_2' & c_2 & & \\ & & b_3' & c_3 & \\ & & & \ddots & \ddots \\ & & & & b_n' \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{Bmatrix} = \begin{Bmatrix} d_1 \\ d_2' \\ d_3'' \\ \vdots \\ d_n^{(n-1)} \end{Bmatrix}$$

Back Substitution:

$$x_n = \frac{d_n^{(n-1)}}{b_n'}$$

Thomas Algorithm

Tridiagonal System:

$$\begin{bmatrix} b_1 & c_1 & & & \\ a_2 & b_2 & c_2 & & \\ & a_3 & b_3 & c_3 & \\ & & \ddots & \ddots & \ddots \\ & & & a_n & b_n \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{Bmatrix} = \begin{Bmatrix} d_1 \\ d_2 \\ d_3 \\ \vdots \\ d_n \end{Bmatrix}$$

Forward Elimination:

$$\begin{bmatrix} b_1 & c_1 & & & \\ & b_2' & c_2 & & \\ & a_3 & b_3 & c_3 & \\ & & \ddots & \ddots & \ddots \\ & & & a_n & b_n \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{Bmatrix} = \begin{Bmatrix} d_1 \\ d_2' \\ d_3 \\ \vdots \\ d_n \end{Bmatrix}$$

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$$\begin{bmatrix} b_1 & c_1 & & & \\ & b_2' & c_2 & & \\ & & b_3' & c_3 & \\ & & & \ddots & \ddots \\ & & & & b_n' \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{Bmatrix} = \begin{Bmatrix} d_1 \\ d_2' \\ d_3'' \\ \vdots \\ d_n^{(n-1)} \end{Bmatrix}$$

Back Substitution:

$$x_n = \frac{d_n^{(n-1)}}{b_n'} \quad \rightarrow \quad x_i = \frac{d_i^{(i-1)} - c_i \cdot x_{i+1}}{b_i'}; \quad i = n-1, n-2, \dots, 1$$

Thomas Algorithm

Example 03-02: Find root of the following system of equations.

$$\begin{bmatrix} 2 & -1 & & \\ -1 & 2 & -1 & \\ & -1 & 2 & -1 \\ & & -1 & 2 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{Bmatrix} = \begin{Bmatrix} 50 \\ 10 \\ 10 \\ 210 \end{Bmatrix}$$

Thomas Algorithm

Example 03-02: Find root of the following system of equations.

$$\begin{bmatrix} 2 & -1 & & \\ -1 & 2 & -1 & \\ & -1 & 2 & -1 \\ & & -1 & 2 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{Bmatrix} = \begin{Bmatrix} 50 \\ 10 \\ 10 \\ 210 \end{Bmatrix}$$

Forward Elimination:

Thomas Algorithm

Example 03-02: Find root of the following system of equations.

$$\begin{bmatrix} 2 & -1 & & \\ -1 & 2 & -1 & \\ & -1 & 2 & -1 \\ & & -1 & 2 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{Bmatrix} = \begin{Bmatrix} 50 \\ 10 \\ 10 \\ 210 \end{Bmatrix}$$

Forward Elimination:

$$\begin{bmatrix} 2 & -1 & & \\ & 1.5 & -1 & \\ & -1 & 2 & -1 \\ & & -1 & 2 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{Bmatrix} = \begin{Bmatrix} 50 \\ 35 \\ 10 \\ 210 \end{Bmatrix}$$

Thomas Algorithm

<u>Example 03-02</u>: Find root of the following system of equations.

$$\begin{bmatrix} 2 & -1 & & \\ -1 & 2 & -1 & \\ & -1 & 2 & -1 \\ & & -1 & 2 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{Bmatrix} = \begin{Bmatrix} 50 \\ 10 \\ 10 \\ 210 \end{Bmatrix}$$

Forward Elimination:

$$\begin{bmatrix} 2 & -1 & & \\ & 1.5 & -1 & \\ & -1 & 2 & -1 \\ & & -1 & 2 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{Bmatrix} = \begin{Bmatrix} 50 \\ 35 \\ 10 \\ 210 \end{Bmatrix} \qquad \begin{bmatrix} 2 & -1 & & \\ & 1.5 & -1 & \\ & & 1.333 & -1 \\ & & -1 & 2 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{Bmatrix} = \begin{Bmatrix} 50 \\ 35 \\ 33.33 \\ 210 \end{Bmatrix}$$

Thomas Algorithm

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Forward Elimination:

$$\begin{bmatrix} 2 & -1 & & \\ & 1.5 & -1 & \\ & -1 & 2 & -1 \\ & & -1 & 2 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{Bmatrix} = \begin{Bmatrix} 50 \\ 35 \\ 10 \\ 210 \end{Bmatrix}
\quad
\begin{bmatrix} 2 & -1 & & \\ & 1.5 & -1 & \\ & & 1.333 & -1 \\ & & -1 & 2 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{Bmatrix} = \begin{Bmatrix} 50 \\ 35 \\ 33.33 \\ 210 \end{Bmatrix}
\quad
\begin{bmatrix} 2 & -1 & & \\ & 1.5 & -1 & \\ & & 1.333 & -1 \\ & & & 1.25 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{Bmatrix} = \begin{Bmatrix} 50 \\ 35 \\ 33.33 \\ 235 \end{Bmatrix}$$

Thomas Algorithm

<u>Example 03-02</u>: Find root of the following system of equations.

$$\begin{bmatrix} 2 & -1 & & \\ -1 & 2 & -1 & \\ & -1 & 2 & -1 \\ & & -1 & 2 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{Bmatrix} = \begin{Bmatrix} 50 \\ 10 \\ 10 \\ 210 \end{Bmatrix}$$

Forward Elimination:

$$\begin{bmatrix} 2 & -1 & & \\ & 1.5 & -1 & \\ & -1 & 2 & -1 \\ & & -1 & 2 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{Bmatrix} = \begin{Bmatrix} 50 \\ 35 \\ 10 \\ 210 \end{Bmatrix} \qquad \begin{bmatrix} 2 & -1 & & \\ & 1.5 & -1 & \\ & & 1.333 & -1 \\ & & -1 & 2 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{Bmatrix} = \begin{Bmatrix} 50 \\ 35 \\ 33.33 \\ 210 \end{Bmatrix} \qquad \begin{bmatrix} 2 & -1 & & \\ & 1.5 & -1 & \\ & & 1.333 & -1 \\ & & & 1.25 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{Bmatrix} = \begin{Bmatrix} 50 \\ 35 \\ 33.33 \\ 235 \end{Bmatrix}$$

Back Substitution:

Thomas Algorithm

<u>Example 03-02</u>: Find root of the following system of equations.

$$\begin{bmatrix} 2 & -1 & & \\ -1 & 2 & -1 & \\ & -1 & 2 & -1 \\ & & -1 & 2 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{Bmatrix} = \begin{Bmatrix} 50 \\ 10 \\ 10 \\ 210 \end{Bmatrix}$$

Forward Elimination:

$$\begin{bmatrix} 2 & -1 & & \\ & 1.5 & -1 & \\ & -1 & 2 & -1 \\ & & -1 & 2 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{Bmatrix} = \begin{Bmatrix} 50 \\ 35 \\ 10 \\ 210 \end{Bmatrix} \quad \begin{bmatrix} 2 & -1 & & \\ & 1.5 & -1 & \\ & & 1.333 & -1 \\ & & -1 & 2 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{Bmatrix} = \begin{Bmatrix} 50 \\ 35 \\ 33.33 \\ 210 \end{Bmatrix} \quad \begin{bmatrix} 2 & -1 & & \\ & 1.5 & -1 & \\ & & 1.333 & -1 \\ & & & 1.25 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{Bmatrix} = \begin{Bmatrix} 50 \\ 35 \\ 33.33 \\ 235 \end{Bmatrix}$$

Back Substitution:

$$x_4 = \frac{235}{1.25} = 188$$

Thomas Algorithm

<u>Example 03-02</u>: Find root of the following system of equations.

$$\begin{bmatrix} 2 & -1 & & \\ -1 & 2 & -1 & \\ & -1 & 2 & -1 \\ & & -1 & 2 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{Bmatrix} = \begin{Bmatrix} 50 \\ 10 \\ 10 \\ 210 \end{Bmatrix}$$

Forward Elimination:

$$\begin{bmatrix} 2 & -1 & & \\ & 1.5 & -1 & \\ & -1 & 2 & -1 \\ & & -1 & 2 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{Bmatrix} = \begin{Bmatrix} 50 \\ 35 \\ 10 \\ 210 \end{Bmatrix} \quad \begin{bmatrix} 2 & -1 & & \\ & 1.5 & -1 & \\ & & 1.333 & -1 \\ & & -1 & 2 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{Bmatrix} = \begin{Bmatrix} 50 \\ 35 \\ 33.33 \\ 210 \end{Bmatrix} \quad \begin{bmatrix} 2 & -1 & & \\ & 1.5 & -1 & \\ & & 1.333 & -1 \\ & & & 1.25 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{Bmatrix} = \begin{Bmatrix} 50 \\ 35 \\ 33.33 \\ 235 \end{Bmatrix}$$

Back Substitution:

$$x_4 = \frac{235}{1.25} = 188$$

$$x_3 = \frac{33.33-(-x_4)}{1.333} = \frac{33.33-(-188)}{1.333} = 166$$

Thomas Algorithm

Example 03-02: Find root of the following system of equations.

$$\begin{bmatrix} 2 & -1 & & \\ -1 & 2 & -1 & \\ & -1 & 2 & -1 \\ & & -1 & 2 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{Bmatrix} = \begin{Bmatrix} 50 \\ 10 \\ 10 \\ 210 \end{Bmatrix}$$

Forward Elimination:

$$\begin{bmatrix} 2 & -1 & & \\ & 1.5 & -1 & \\ & -1 & 2 & -1 \\ & & -1 & 2 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{Bmatrix} = \begin{Bmatrix} 50 \\ 35 \\ 10 \\ 210 \end{Bmatrix} \quad \begin{bmatrix} 2 & -1 & & \\ & 1.5 & -1 & \\ & & 1.333 & -1 \\ & & -1 & 2 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{Bmatrix} = \begin{Bmatrix} 50 \\ 35 \\ 33.33 \\ 210 \end{Bmatrix} \quad \begin{bmatrix} 2 & -1 & & \\ & 1.5 & -1 & \\ & & 1.333 & -1 \\ & & & 1.25 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{Bmatrix} = \begin{Bmatrix} 50 \\ 35 \\ 33.33 \\ 235 \end{Bmatrix}$$

Back Substitution:

$$x_4 = \frac{235}{1.25} = 188$$

$$x_3 = \frac{33.33-(-x_4)}{1.333} = \frac{33.33-(-188)}{1.333} = 166$$

$$x_2 = \frac{35-(-x_3)}{1.5} = \frac{35-(-166)}{1.5} = 134$$

Thomas Algorithm

Example 03-02: Find root of the following system of equations.

$$\begin{bmatrix} 2 & -1 & & \\ -1 & 2 & -1 & \\ & -1 & 2 & -1 \\ & & -1 & 2 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{Bmatrix} = \begin{Bmatrix} 50 \\ 10 \\ 10 \\ 210 \end{Bmatrix}$$

Forward Elimination:

$$\begin{bmatrix} 2 & -1 & & \\ & 1.5 & -1 & \\ & -1 & 2 & -1 \\ & & -1 & 2 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{Bmatrix} = \begin{Bmatrix} 50 \\ 35 \\ 10 \\ 210 \end{Bmatrix} \quad \begin{bmatrix} 2 & -1 & & \\ & 1.5 & -1 & \\ & & 1.333 & -1 \\ & & -1 & 2 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{Bmatrix} = \begin{Bmatrix} 50 \\ 35 \\ 33.33 \\ 210 \end{Bmatrix} \quad \begin{bmatrix} 2 & -1 & & \\ & 1.5 & -1 & \\ & & 1.333 & -1 \\ & & & 1.25 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{Bmatrix} = \begin{Bmatrix} 50 \\ 35 \\ 33.33 \\ 235 \end{Bmatrix}$$

Back Substitution:

$$x_4 = \frac{235}{1.25} = 188$$

$$x_3 = \frac{33.33-(-x_4)}{1.333} = \frac{33.33-(-188)}{1.333} = 166$$

$$x_2 = \frac{35-(-x_3)}{1.5} = \frac{35-(-166)}{1.5} = 134$$

$$x_1 = \frac{50-(-x_2)}{2} = \frac{50-(-134)}{2} = 92$$